Types of Porphyria Cutanea Tarda (PCT)

Acquired (Type I) PCT

In acquired (type I) PCT, uroporphyrinogen (UROGEN) decarboxylase deficiency is restricted to the liver. This tissue-specific deficiency may result from either a liver-specific gene defect or exposure to chemicals that selectively inhibit the hepatic form of the enzyme without affecting the erythrocyte (RBC) enzyme. Certain substances trigger PCT only in susceptible individuals—such as **alcohol** and **estrogens**—while others, like **hexachlorobenzene (HCB)**, can induce the disease in nearly all exposed individuals.

Hereditary (Type II) PCT

Hereditary (type II) PCT is an autosomal dominant disorder characterized by approximately 50% reduction in UROGEN decarboxylase activity across all tissues, including RBCs and cultured skin fibroblasts. Enzyme levels show a bimodal distribution: - A large group (>80%) has normal enzyme concentration. - A smaller group (<20%) exhibits about half the normal amount of detectable enzyme.

Some patients with type II PCT have enzyme activity at the lower end of the normal range. Clinical penetrance is low (~20%), indicating that most individuals with the genetic defect do not develop symptoms. This implies that additional genetic or environmental factors are required for disease expression. Over 100 mutations in the *UROGEN decarboxylase* gene have been identified, many of which impair enzyme stability or cause defective pre-mRNA splicing.

Type III PCT

Not all patients with a family history of PCT have type II disease. Some individuals have multiple affected relatives but normal RBC UROGEN decarboxylase activity. This variant has been termed **type III PCT** by some researchers. Possible explanations include: - Inheritance of a UROGEN decarboxylase variant that appears normal immunologically but is selectively vulnerable to hepatic inhibition. - Presence of a second inherited enzyme defect unrelated to UROGEN decarboxylase.

These hypotheses remain under investigation.

Differential Diagnosis and Dual Porphyrias

Some cases originally classified as hereditary PCT may actually represent **variegate porphyria (VP)**, another autosomal dominant porphyria. Families with both PCT and VP have been described ("dual porphyrias"). Additionally, coexistence of PCT and **acute intermittent porphyria (AIP)** has been reported, further complicating diagnosis and classification.

Precipitating Factors in Acquired (Type I) PCT

Multiple agents and conditions are associated with the development of sporadic (type I) PCT:

- **Alcohol**
- **Estrogenic hormones**
- **Iron overload**
- **Viral infections** (hepatitis C, HIV)
- **Polychlorinated hydrocarbons** (e.g., HCB, TCDD)
- **Hemodialysis** in renal failure

Alcohol

Alcohol consumption is a well-established trigger for PCT. Ethanol induces hepatic **ALA synthase**, the rate-limiting enzyme in heme synthesis. It also reduces erythrocyte UROGEN decarboxylase activity in both acute and chronic alcohol use. Ethanol inhibits other heme pathway enzymes, including **ferrochelatase** and **ALA dehydratase**. Chronic alcoholism suppresses erythropoiesis and increases intestinal iron absorption, effects potentially amplified by mutations linked to **hemochromatosis**.

Table 132-6: Drugs and Chemicals Associated with PCT

- Ethyl alcohol
- Estrogenic hormones
- Hexachlorobenzene (HCB)
- Chlorinated phenols
- Iron
- 2,3,7,8-Tetrachlorodibenzo-*p*-dioxin (TCDD)
- Polychlorinated biphenyls (PCBs)
- Herbicides: 2,4-dichloro- and 2,4,5-trichlorophenoxyacetic acid